DIY Jon-Air Airboat: Build Guide

Version 1.0.1.3, minimal 1448 build, Rotax 800R E-TEC. All costs and hours are estimates. Each step and section shows what it costs if you do it yourself (materials and DIY hours) or hire it out (pro cost, at $85/h, on top of materials). Done entirely DIY, the build is about $16,270 and 215 hours. With every step hired out, it is about $24,983. Steps with 0 pro hours, such as buying the boat and choosing the prop, are yours to do either way. Generated from the Build Guide tab of the tracker; BOM line numbers refer to that tracker.

	Materials	DIY hours	Pro hours	Pro cost if hired	Total if hired
Section 1 - Hull Armor	$4,735	28	9	$765	$5,500
Section 2 - Install Drivetrain And Prop Cage	$11,255	181	89.5	$7,608	$18,862
Section 3 - Static Test	$280	6	4	$340	$620
Total	**$16,270**	**215**	**102.5**	**$8,712**	**$24,983**

Section 1 - Hull Armor

	Materials	DIY hours	Pro hours	Pro cost if hired	Total if hired
Section total	**$4,735**	**28**	**9**	**$765**	**$5,500**

1.1 Buy a used 1448 jonboat

	Materials	DIY hours	Pro hours	Pro cost if hired	Total if hired
Step 1.1	$4,000	10	0	$0	$4,000

Depends on: - | Alongside: -

1. Raise the purchase money (see BOM line 1 for the current price estimate).
2. Search Facebook Marketplace, Craigslist (Orange County, LA, Inland Empire, Ventura) and OfferUp for "1448", "14 ft jon boat", "jonboat". Set a saved search / alert.
3. Target: 14 ft long, 48 in bottom width, riveted or welded aluminum, flat bottom with ribs inside. Lowe, Tracker, Alumacraft, Crestliner, Polar Kraft are all fine. A trailer is a big plus.
4. Before visiting, ask for: photos of the bottom and transom, title / registration (CA "pink slip") in the seller's name, and the HIN (hull ID number) on the transom.
5. Inspect in person: look for cracks around the transom and ribs, corrosion (white powder, pitting), dents with creases, and rivets with a dark halo (weeping). Ask to put a few inches of water in the hull (or look at it on the water) and check underneath for drips.
6. Negotiate. If the boat comes with a motor you don't need, factor in its resale value.
7. Pay, get the signed title and a bill of sale, and transfer registration at the DMV within 10 days.
8. Tow it home (check that trailer lights and tires work) or arrange transport.

Skills: Fundraising; Use of online 'for sale' forums

1.2 Add UHMW sheet to the bottom of the hull

	Materials	DIY hours	Pro hours	Pro cost if hired	Total if hired
Step 1.2	$735	18	9	$765	$1,500

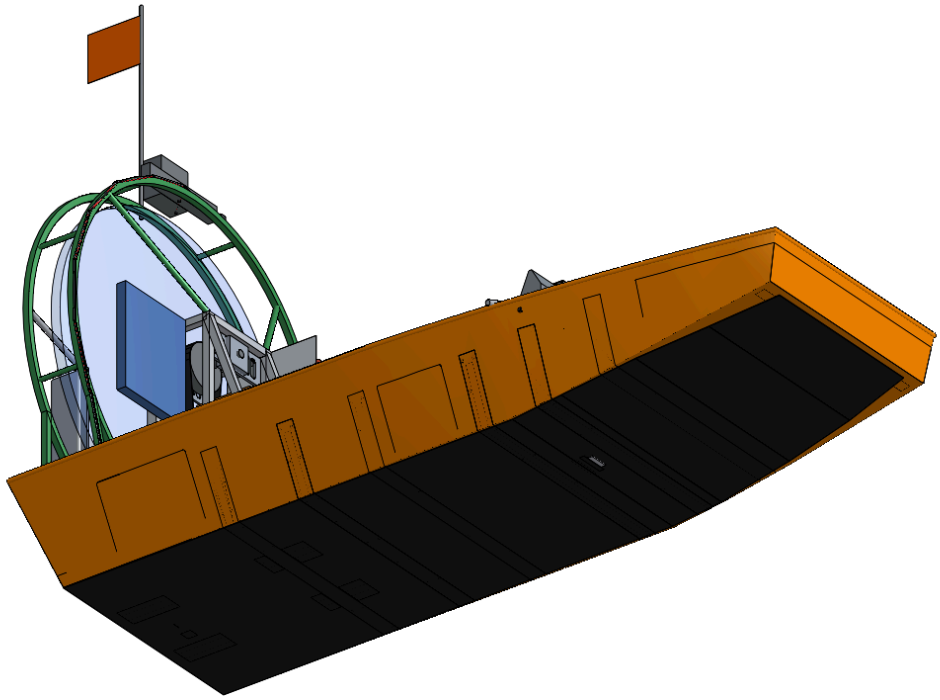
Depends on: 1.1 | Alongside: 2.1, 2.8 (shop work)

MINIMAL BUILD: 1/8 in UHMW over the whole bottom from the transom to the bow, about 36 lb with fasteners and backing bars. Two 4 x 8 sheets end to end, joined on the rib at X 95.25. The 48 in sheet leaves about 0.4 in of bare hull along each chine, and the bottom narrows forward of about X 112, so trim to fit there. 1/8 in is thin: it wears through sooner than 3/16 in, so look it over each season and replace a worn piece.

Cut list: aft piece, one full sheet, cut to length. Forward piece, about 48 x 62 in, from the second sheet. Cut it 2 in long and trim it to the bow on the hull. Windows: 3 x 12 in, one under each of the four shoes (X 2 to 13 and X 34 to 45, centered on Y +/-8.5), so the shoe bolt heads in step 2.6 seat on the hull and the sheet can still slide.

1. Flip the hull upside down onto sawhorses or old tires with a helper (about 150 lb). Never work under a hull that is not blocked securely.
2. Clean the bottom: pressure wash, scrape off growth, degrease with acetone. Leak-test first and fix any weeping rivets now; the UHMW will hide them.
3. From inside the hull, measure each rib's distance from the transom and mark it on the outside (approx. 37.9, 54.75, 70.75, 95.25, 108.25 and 122.25 in on the reference hull). Add three rows where there is no rib: X 3, X 140 and X 155. Snap a centerline. Mark eight screws across each row, about 6 in apart, at least 1 in clear of the windows.
4. Check you can reach the X 140 and X 155 rows from inside, under the front shelf, before cutting anything. Cut two backing bars, 1 x 1/8 in 6061, about 44 in long, and trim them to fit. They stand in for a rib. If you can't reach a row, stop the sheet at the last row you can reach.
5. Cut the sheets with a circular saw (fine-tooth plywood blade) against a straightedge. Cut the windows. Round every corner, and chamfer the forward edge and both chine edges with a file or router.
6. Work at a moderate temperature (60-80 F). Leave 1/4 in at the joint; UHMW grows and shrinks about 10 times as much as aluminum.
7. Clamp the forward piece first, working from the rib at X 95.25 up the bow rise. 1/8 in bends cold. At each row, drill a 3/16 in pilot through UHMW and hull, on the rib flange or backing bar and at least 1/2 in from existing rivets. Row X 3 has no flange: use fender washers on the skin. Then fit the aft piece the same way, with a second row just aft of the joint.
8. Open all holes to #7 (13/64 in). Fixed row = the forward row of each piece: leave round. All other rows: slot the hole in the UHMW only, about 5/8 in long fore and aft (drill two holes and file between them). The hull hole stays round.
9. Countersink the UHMW with the 100 deg bit until the screw head sits flush. Test on an offcut first: an 82 deg head is about as deep as this sheet, so don't use one. Stop before the cut reaches the aluminum.
10. Remove the sheet, deburr the aluminum holes, and wipe with acetone.
11. Reinstall the sheet. Put a dab of 4000UV in each hull hole and on each screw under the head. Insert #10-24 x 1 in flat-head screws from outside. Run a bead of sealant along the forward edge at X 157.
12. Inside the hull, put a fender washer and a dab of sealant on the rib flange or backing bar, then the nylock nut. A helper holds the screwdriver outside. Tighten the fixed rows snug. Tighten slotted rows just until the head seats, without crushing the UHMW, so the sheet can move.

13. Repeat for every row, about 80 screw sets in all, wiping off excess sealant. Let cure 24 h.
14. Flip the hull back over. Water-test (a few inches of water in the hull, or launch) and check every bolt from outside and inside for drips. Re-bed any that weep.



Skills: Measuring & layout; Power drill; Circular saw (straight cuts); Countersinking; Applying marine sealant; Basic hand tools (wrench, screwdriver); Safe lifting/flipping with a helper

Section 2 - Install Drivetrain And Prop Cage

	Materials	DIY hours	Pro hours	Pro cost if hired	Total if hired
Section total	**$11,255**	**181**	**89.5**	**$7,608**	**$18,862**

MINIMAL BUILD: a Rotax 800R E-TEC snowmobile two-stroke driving the prop through one 8MGT belt, on a bolted aluminum stand, with a smaller aluminum cage. The rear bench stays. The middle bench is replaced by the seat module (2.17): two struts across the boat, an electrical dry box, dry storage each side, a closed-cell foam seat for two and a folding seatback. Design details are in reports/Drivetrain_800R_ETEC.md; wiring is in electrical/Minimal_1448_Electrical_Schematic.pdf and electrical/Wire_List.csv.

Dimensions use the CAD frame: X forward from the transom, Y + to port, Z up from the hull bottom, all in inches. They come from a reference model of a Lowe 1448. Check them on your own hull, and stop if anything differs by more than 1/2 in.

Approach: build the stand and drivetrain as a module off the boat and lower it onto waiting shoes. Then connect the systems, build the cage around the real prop position, and fit the prop last.

Step 2.16 can be done any time after 2.13. Step 2.17 is listed last but is done right after 2.5: its struts replace the bench as the thwart, and the electrical work in 2.11 needs its dry box. Step 2.18 (oarlocks) can be done any time after 2.17. Finish all three before the sign-off in 2.14.

2.0 Choose your prop (standard 3-blade, or the quieter 4-blade)

	Materials	DIY hours	Pro hours	Pro cost if hired	Total if hired
Step 2.0	$0	1	0	$0	$0

Depends on: - | Alongside: 1.1, 1.2

Decide this before building the belt reduction (step 2.8): the prop sets the large pulley size.

STANDARD (this guide and the BOM): Warp Drive 3-blade airboat prop with HP blades, cut to 66 in (BOM line 32, $1,275 plus face plate and flange). Belt 8MGT 32T / 94T. Prop about 2,590 rpm at 7,600 engine rpm, tip speed about 745 ft/s.

QUIETER OPTION: Warp Drive 4-blade with HP blades, cut to 66 in ($1,680, about $405 more, plus about $30 for a larger pulley). It turns about 10% slower for the same power: about 2,340 rpm, tip speed about 675 ft/s. Thrust and top speed are about the same; it runs smoother and is noticeably quieter for the operator and from shore: about 3-4 dBA less prop noise at 50 ft (estimate). The total gain depends on the exhaust: with a loud exhaust it is only about 1-1.5 dBA, so fit a good muffler either way. Neither version is expected to meet the California shoreline limit (75 dB(A), SAE J1970): estimates are about 82-84 dBA (3-blade) and 79-82 dBA (4-blade) at 50 ft. Florida's airboat law (s. 327.391) requires a muffler but gives no decibel figure in the statute. The stock silencer is the muffler here.

IF YOU CHOOSE THE 4-BLADE, change these when you build:

1. BOM line 32: order the 4-blade (Warp Drive HP blades, 4-blade, up to 74 in, $1,680), cut to 66 in, with the same SAE #2 face plate and flange.
2. BOM line 31: order the large (prop shaft) pulley as 8MGT 104T instead of 94T. The small (crank PTO) pulley stays 32T.
3. Belt length: with the 52T pulley the belt is longer for the same center distance (about 30 in). Recalculate the belt length in Gates DesignFlex (or the pulley maker's calculator) before ordering, and check that the jack-bolt slots in the bearing plates still give tensioning travel both ways.
4. Clearance: the 104T pulley is about 1 in larger in diameter (10.43 in pitch diameter against 9.42 in): check its clearance to the stand, the engine and the cage.
5. Loads: prop torque rises about 11% (to about 270 lb-ft) and the turning (gyroscopic) load about 20%. The 1-1/2 in shaft and the bearings still have margin. To recheck, move the 104T belt to the top of the BELTS list in CAD/analysis/etec_reduction.js and run it.
6. Weight: about 3-4 lb more.
7. Pitch setting (both props): start at the prop maker's suggested pitch, do a tied-down full-throttle run, and adjust the pitch until the engine reaches 7,400 to 7,500 rpm (never over its 7,900 rpm maximum). Never exceed the maker's 3,200 prop rpm or 160 hp. Get Warp Drive to approve this engine and belt combination before ordering.

Everything else in the build (stand, cage, prop shaft, bearings, flange) is the same for both props.

Skills: Reading a spec sheet; comparing options

2.1 Verify the layout on your boat and make templates

	Materials	DIY hours	Pro hours	Pro cost if hired	Total if hired
Step 2.1	$60	8	4	$340	$400

Depends on: 1.1 | Alongside: 1.2 (if not yet done)

Goal: confirm the design fits your hull before cutting anything.

1. Measure from the transom and centerline: rib positions, rear bench (about X 13-33, top about Z 13.7), middle bench (about X 75.5-89.5, top about Z 12.5), gunwale height, transom height.

2. Check what the design depends on:

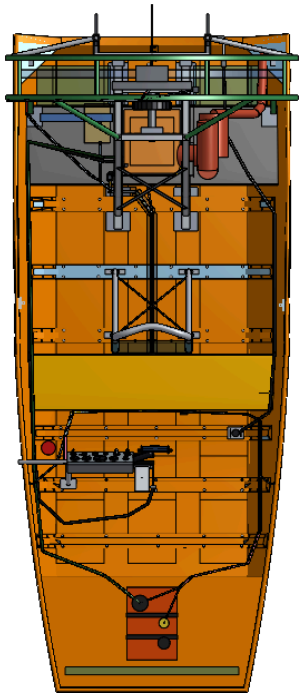
- Bench panels: their gauge, how they fasten to the hull, and whether they stop at about Z 2.4.
- Corner caps: whether they are aluminum and strong enough to carry a cage leg.
- Rear bench: confirm it is full of sound foam, because the capsize flotation in step 2.16 relies on it, and that the access pocket is practical.
- Middle bench: it comes out in 2.5. Note how it is fastened to the hull sides, ribs and floor, and measure the beam across the gunwales at the bench.
- Rib height at Y +/-8.5, and the hull bottom gauge.
- Transom condition around Z 4.5, where the clips bolt on.
- Front deck: strength under 45 lb of fuel, and how to fasten the straps.
- Rib height where the stick base and console foot land.

3. Weigh and measure the engine, tuned pipe, silencer, oil tank and radiator. Update the layout if they differ from the CAD envelopes, and choose the engine roll angle so the stock pipe fits. The engine sits about X 14-30.

4. Cut 1/4 in plywood templates for the four shoes, four cage foot pads, and the dry box, battery switch and harness sleeve once the seating redesign sets their places.

5. Lay them out in the boat and write the measured numbers on them. Photograph everything.

If anything differs from the CAD by more than 1/2 in, stop and update the CAD before going on.



Skills: Measuring and layout; Reading the CAD model; Basic woodworking (templates)

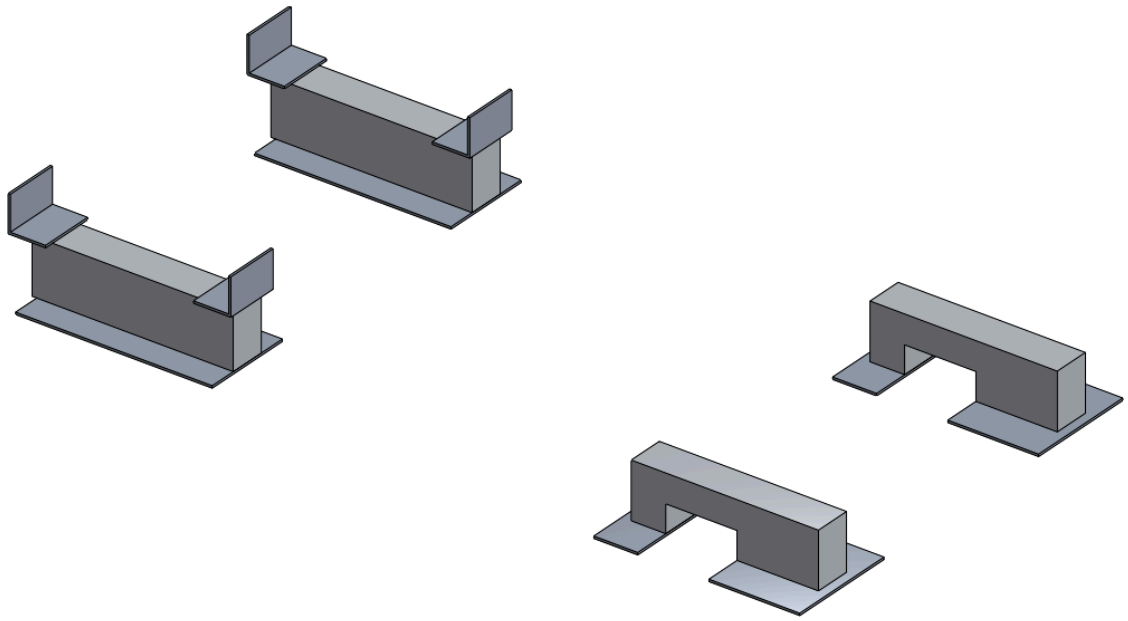
2.2 Fabricate the shoes, doublers and clips (shop)

	Materials	DIY hours	Pro hours	Pro cost if hired	Total if hired
Step 2.2	$54	6	3	$255	$309

Depends on: 2.1 | Alongside: 2.3, 2.4, 2.5, 2.8

Goal: four separate shoes, not continuous beds, so the rear bench and its flotation stay uncut.

1. Aft shoes (2): 2 x 3 x 1/8 in 6061 tube, X 2-12.75, at Y +/-8.5.
2. Forward shoes (2): the same tube, X 34-44, notched over the rib at X 36-39.5. Leave about 0.08 in over the rib top, so a hard landing bears on the rib.
3. Doublers, 1/8 in 5052: 5 x 10.5 in under each aft shoe. Under each forward shoe, two 6 in wide pieces, cut around the rib.
4. Clips: 2 x 2 x 1/8 in angle, 3 in wide. One at each end of each aft shoe, to the transom and to the rear bench's aft panel.
5. Drill for six 1/4-20 bolts per shoe. Cut a crush sleeve for each bolt, so the bolts can be tightened without collapsing the tube.
6. Deburr everything. Round all corners.



Skills: Cutting aluminum; Drilling; Layout from templates

2.3 Cut and drill the stand side frames (shop)

	Materials	DIY hours	Pro hours	Pro cost if hired	Total if hired
Step 2.3	$65	8	4	$340	$405

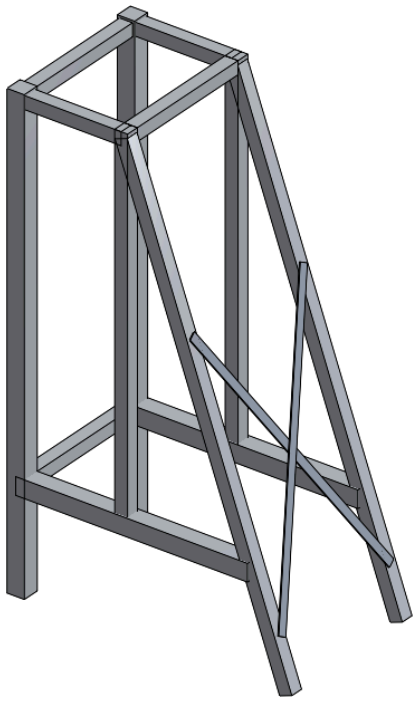
Depends on: 2.1 | Alongside: 2.2, 2.4, 2.5, 2.8

Goal: all stand parts cut and drilled, ready to bolt together in 2.8. No welding is needed.

Side frames at Y +/-8.5, 6061, 1/8 in wall:

1. Aft posts: 2 x 2 in, X 9, from the shoe to Z 57.
2. Rails: 2 x 2 in, Z 14.9, from X 9 to about X 36. They clear the bench top by only 0.2 in; add a rubber strip if they touch under load.
3. Forward posts: 1-1/2 x 1-1/2 in, X 21, from the rail to Z 56.75.
4. Top rails: 1-1/2 x 1-1/2 in, Z 56, X 9-21.
5. Diagonals: 1-1/2 x 1-1/2 in, from the top of the forward post down to the forward shoe at X 40.
6. Ties across the boat: 1-1/2 x 1-1/2 in, at X 9 Z 14.9, X 9 Z 56 and X 21 Z 56.
7. Cross-straps: two 1 x 1/8 in flats in an X on the forward face of the diagonals, below the cylinders.
8. Gussets: 1/4 in 6061 at the joints, with four 5/16 in bolts on each side of each joint. Drill them as templates so both side frames match.

The rail is the most heavily loaded member, with a load factor of about 1.4 assuming engine mounts at X 17 and X 27. If your real mounts differ, recheck it with CAD/analysis/stand_800r.js.



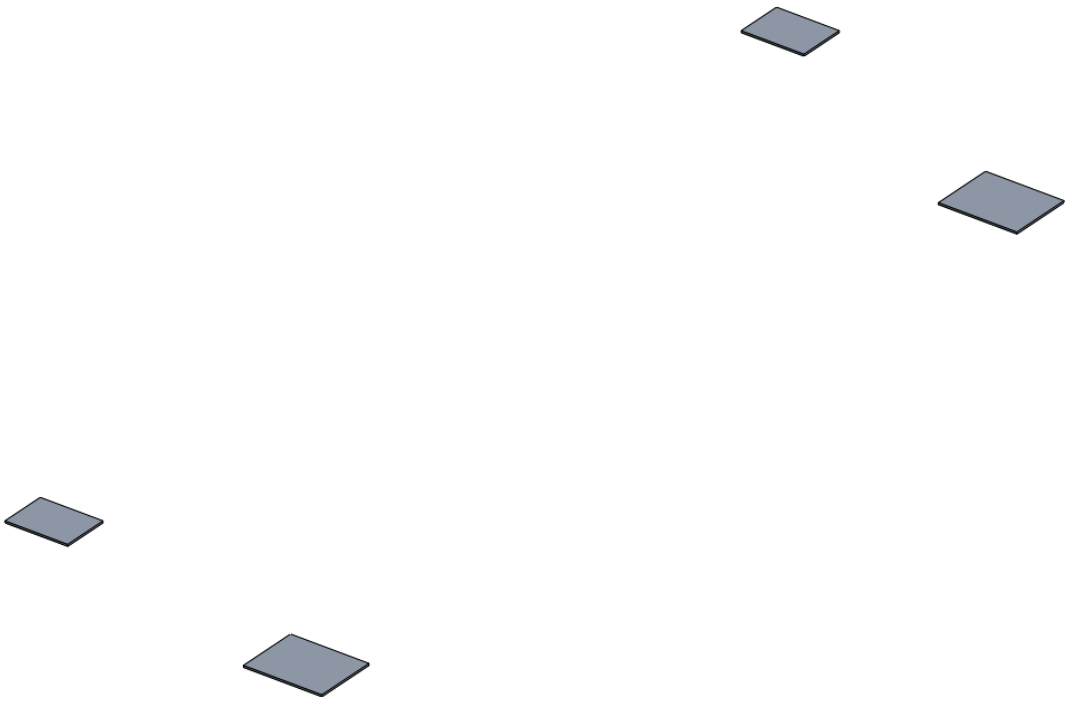
Skills: Cutting aluminum; Accurate drilling; Layout; Reading drawings

2.4 Fabricate cage foot pads and backing plates (shop)

	Materials	DIY hours	Pro hours	Pro cost if hired	Total if hired
Step 2.4	$0	3	1.5	$128	$128

Depends on: 2.1 | Alongside: 2.2, 2.3, 2.5, 2.8

1. Aft pads (2): 1/8 in 6061, 4 x 3 in, for the transom corner caps, at about X -1 to 3 and Y +/-29.75 to 32.75.
2. Forward pads (2): 1/8 in 6061, 5 x 4 in, for the rear bench top, at about X 13.25-18.25 and Y +/-27.5 to 31.5.
3. A matching backing plate for each pad.
4. Drill four 1/4-20 holes through each pad and backing plate as a pair, and mark them so they stay matched.
5. Round the corners and deburr.



Skills: Cutting aluminum; Drilling; Layout

2.5 Remove the middle bench; open the rear bench access pocket (boat)

	Materials	DIY hours	Pro hours	Pro cost if hired	Total if hired
Step 2.5	$105	5	2.5	$212	$318

Depends on: 2.1 | Alongside: 2.2, 2.3, 2.4, 2.8

Goal: the middle bench out of the boat, every hole it leaves sealed, and the hull held to its measured beam until the new struts go in. The new seat, electrical dry box and storage compartments replace it (seating redesign, v1.0.1.3).

Middle bench (about X 75.5-89.5, top about Z 12.5):

1. Before anything else, measure and photograph: the beam across the gunwales at the bench, and every rivet and bracket that holds the bench to the hull sides, ribs and floor. The new gunwale struts and dry box use the same locations.
2. Clamp a temporary spreader across the gunwales at the bench (a 2x4 cut to the measured inside beam, padded at the ends). The bench is the boat's main thwart: leave the spreader in until the new struts are bolted in, and do not lift, trailer or launch the boat without it.
3. Drill out the rivets that hold the bench: use a bit the size of the rivet shank, only through the head, then punch the shank out. Do not enlarge the holes in the hull skin.
4. Lift the bench out whole, with its foam. Record the foam volume (about 14 x 61 x 11 in of bench, less the panels): the capsize flotation relied on it, and the seating redesign restores it.
5. Seal every empty hole: the bench ends sit low on the hull sides, near or below the waterline. Fill each with a 3/16 in closed-end (sealed) aluminum rivet set in marine sealant, unless the new struts will reuse it; tape the ones to be reused.
6. Clean the old sealant and any corrosion from the bench footprint. Do not cut or drill for the dry box, battery switch or harness sleeve here: their new locations come with the seating redesign (BOM lines 57 and 76

and the harness sleeve in line 73 are pending it).

Rear bench:

7. Cut a small access pocket through the aft panel into the foam, so you can reach the forward cage pad backing plates. Refill and seal it after 2.7.
8. Seal all exposed rear bench foam. Restore any lost rear bench flotation with closed-cell pour foam.

Skills: Drilling out rivets; Riveting; Measuring; Foam work; Marine sealant

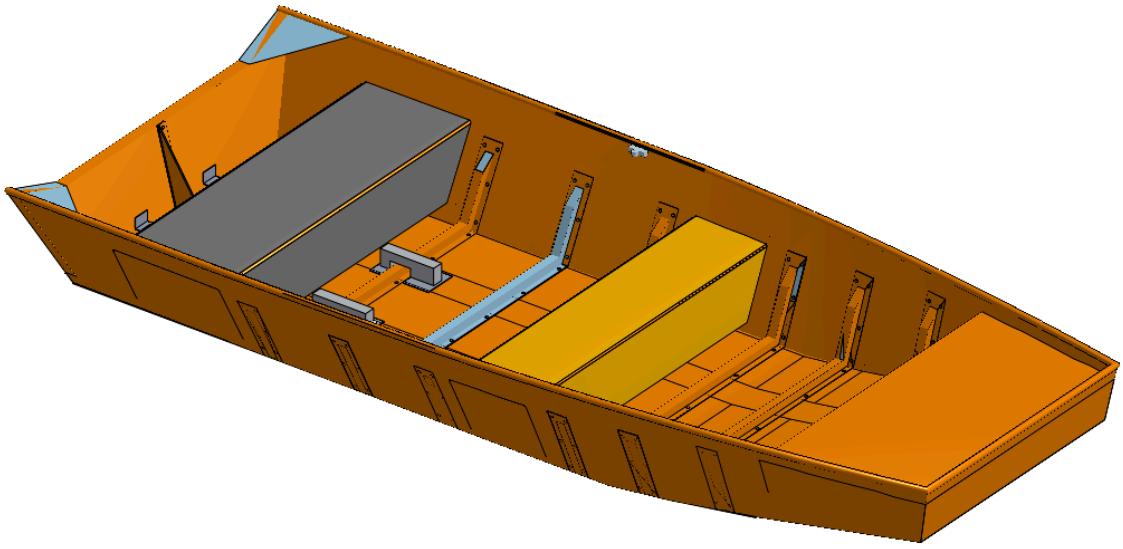
2.6 Install the shoes and doublers (boat)

	Materials	DIY hours	Pro hours	Pro cost if hired	Total if hired
Step 2.6	$144	8	4	$340	$484

Depends on: 2.2, 2.5 | Alongside: 2.4, 2.8

Goal: four shoes tied to the bottom, the rib, the transom and the bench panel.

1. Dry-fit all four shoes on their doublers. Check they are parallel, at Y +/-8.5, and level across the boat.
2. Drill through shoe, doubler and bottom from inside. Fit the crush sleeves.
3. Bolt with six 1/4-20 stainless bolts per shoe and sealant in every hole. Use truss heads outside. The bolts pass through the window cut in the UHMW sheet in step 1.2, so the heads seat on the hull.
4. Aft shoes: fit the clips to the transom (about Z 4.5) and to the rear bench's aft panel. These carry the uplift: about 394 lbf per side at full power, so don't skip them.
5. Forward shoes: check the rib notch. The shoe should nearly touch the rib top, with about 0.08 in of clearance.
6. Water-test the bottom and fix any weeps before going on.



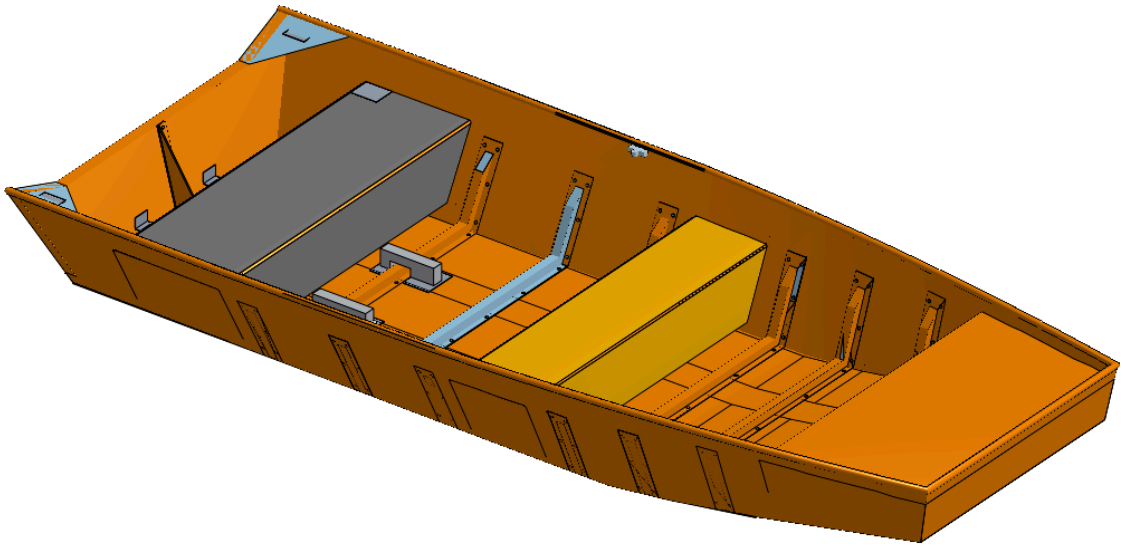
Skills: Drilling aluminum; Measuring and leveling; Bolting with a helper; Marine sealant

2.7 Install the cage foot pads (boat)

	Materials	DIY hours	Pro hours	Pro cost if hired	Total if hired
Step 2.7	$35	4	2	$170	$205

Depends on: 2.4, 2.5 | Alongside: 2.6, 2.8, 2.9

1. Aft pads: bolt to the transom corner caps with backing plates and four 1/4-20 bolts each, at about Z 17.9.
2. Forward pads: bolt to the rear bench top next to the hull side, with backing plates reached through the access pocket from 2.5.
3. Check that the pads are level across the boat and at the right height for the cage legs.
4. Refill and seal the access pockets.
5. Water-test.



Skills: Drilling aluminum; Measuring and leveling; Bolting with a helper; Marine sealant

2.8 Build the stand and drivetrain module (shop)

	Materials	DIY hours	Pro hours	Pro cost if hired	Total if hired
Step 2.8	$3,749	28	14	$1,190	$4,939

Depends on: 2.1, 2.3 | Alongside: 2.2, 2.4-2.7 (biggest parallel task)

Goal: the whole drivetrain built and checked on the floor as one liftable unit.

Stand:

1. Use a plywood jig with the shoe spacing. Bolt up both side frames, ties, gussets and cross-straps.

Bearing plates (BOM line 31, cut by a waterjet or CNC shop):

2. Aft plate 3/8 in 6061 at X 8, forward plate 1/2 in 6061 at X 20. Each has two petals either side of the bearing, a 1/2 in rib, a 1 in border, and solid ends with the tensioning slots.

3. Round and polish every cut edge. Tool marks are where fatigue cracks start. Anodize if you want the finish.
4. Bolt the plates between the side frames, with jack bolts for belt tension.

Prop shaft:

5. 1-1/2 in 4140 shaft on a tapered-roller flange bearing forward (carries the thrust) and a UCF208-24 ball bearing aft. Fit two clamp collars either side of the forward bearing's inner ring. The set screws only locate it.
6. Fit the 94T pulley (104T for the 4-blade) at about X 13. Both pulleys must be flanged and dynamically balanced.

Engine:

7. Confirm the PTO taper and side load from the service manual before machining the 32T hub. If the manual can't confirm the side load, add an outboard support bearing.
8. Mount the engine on the sled rubber mounts, crank along X, PTO aft, crank at Z 22.72, 0.72 in higher than the original layout. That gives 29.28 in centers for the 8MGT-2000-36 belt.
9. Fit the belt. Align the pulleys with a straightedge and keep the shafts parallel. Tension a new belt with the jack bolts to 457 to 499 lb static, or 33 to 36 lb at mid-span to deflect the span 0.45 in. Lock the jack bolts, then fit the belt guard. A wrong-tension belt fails early.

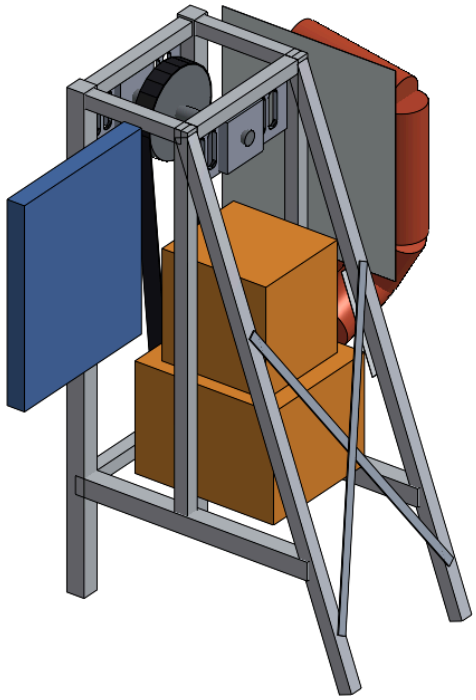
Attached to the stand:

10. Oil tank, upright, on the starboard side frame, X 12.6-22.4, Z 17-26, filler on top. Keep the sled oil level sensor.
11. Radiator, 18 x 22 x 2 in, starboard, X 15-17, Y -28 to -10, Z 34-56, with the overflow bottle. It must see clean air, never air from behind the exhaust.
12. Tuned pipe and stock silencer on the port side, unmodified. Keep the sled springs and ball joint, and use a flexible joint between engine-mounted and stand-mounted parts.
13. Heat shields on standoffs between the exhaust and the stand, belt and bearings. Shields, not wrap.

Finish:

14. Add two lifting points at the balance point.
15. Torque every fastener and paint a torque stripe across each nut.
16. Weigh the module and write the weight on the frame. Update CAD/analysis/minimal_1448.js. The engine's 79 lb is unverified.

The prop is NOT fitted here (see 2.15).



Skills: Mechanical assembly; Engine mechanics; Torque wrench use; Reading drawings; Belt tensioning

2.9 Rig the lift and do a trial lift (shop)

	Materials	DIY hours	Pro hours	Pro cost if hired	Total if hired
Step 2.9	$95	3	1.5	$128	$222

Depends on: 2.8 | Alongside: 2.7

1. Set up an engine hoist or gantry rated for at least twice the module weight, with rated slings and shackles.
2. Lift 2 in off the floor and hold for 5 minutes. Check balance, that the stand does not rack, and that nothing touches the slings.
3. Plan the path into the boat: boat on its trailer, chocked, tongue on a stand, area clear.

Never put any part of your body under a suspended load.

Skills: Rigging and safe lifting; Engine hoist operation

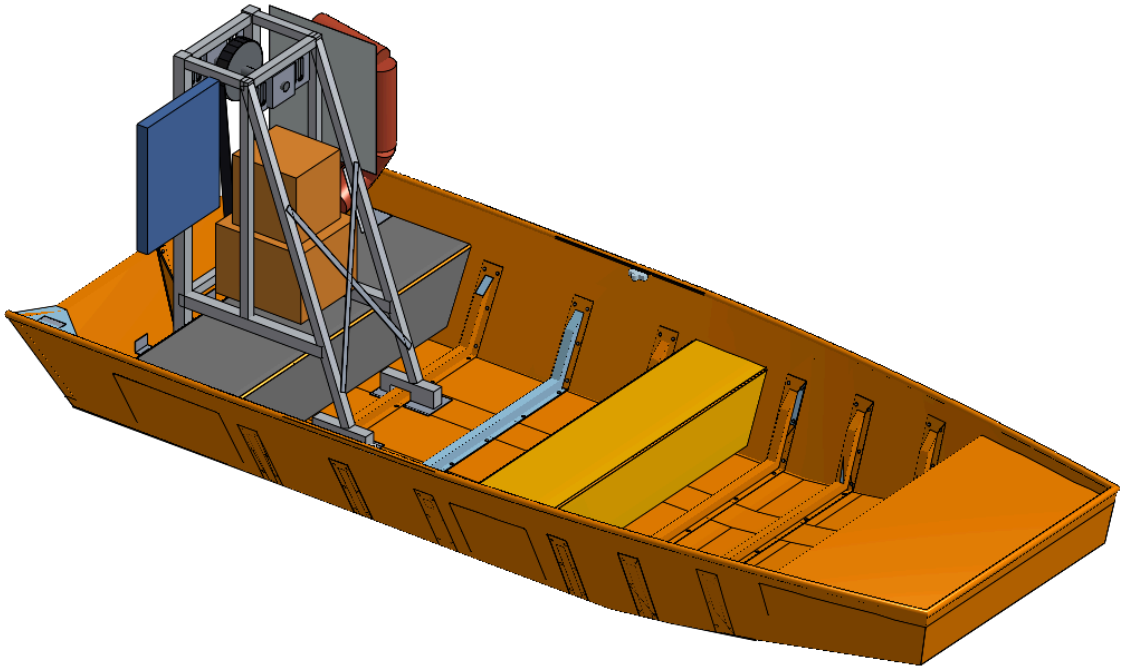
2.10 Lower the module into the boat and bolt it down (boat, 3 people)

	Materials	DIY hours	Pro hours	Pro cost if hired	Total if hired
Step 2.10	$0	4	2	$170	$170

Depends on: 2.6, 2.9 | Alongside: -

1. Lift and move slowly over the boat, with one person on the hoist and two guiding.
2. Lower the rails over the rear bench. They clear the bench top by only 0.2 in, so go slowly.
3. Land the aft posts on the aft shoes and the diagonals on the forward shoes. Line up the holes with drift pins.
4. Bolt down with threadlocker. Torque and stripe.
5. Check that nothing touches the benches, hull or ribs, and that the engine mounts aren't bottomed.

6. Check the module against the hull: engine to stand about 0.75 in, and the pipe's 4-6 in from the belt and bearings.



Skills: Rigging; Teamwork (3 people); Torque wrench use

2.11 Connect electrical, fuel, oil, cooling, exhaust and controls (boat)

	Materials	DIY hours	Pro hours	Pro cost if hired	Total if hired
Step 2.11	$3,238	48	24	$2,040	$5,278

Depends on: 2.10, 2.17 | Alongside: 2.12, 2.13

Do the wiring with the battery disconnected. Work from the schematic and wire list; each wire has a number, so label both ends.

Cockpit:

1. Dry box (built in 2.17): battery strapped on its pad, fuse block on its bracket over the battery with the fuses facing the lid, relays K1-K5 on their strip.
2. Battery switch on its plate on the forward seat strut, port of the box, knob forward; its cables enter the box through the forward gland.
3. Console: foot plate on the rib at X 108, post, and arm to the gunwale. Mount the pod, gauges, switches and phone holder. Drill the low oil LED between the temperature and exhaust gauges.
4. Fuel tank on the front deck step, X 134-152, strapped down, with the sled pump module and gauge sender.
5. Fire extinguisher upright on the starboard footwell foam block, by the driver's right knee, about X 99.5, Y -25, in a bracket strapped to the block and the hull side. Air horn in its clip.

Wiring:

6. Main power: W57 from the battery, with the 6 AWG fusible link, to the switch; W01 from the switch to the starter solenoid; W03 from the switch through MF1 to the hub; W02 and W04 for the negative side.

7. F7 and the bilge pump feed are inline at the battery, ahead of the switch, so the automatic pump always has power.
8. Harness forward: out through the gland in the dry box forward wall (Y 3.8, Z 8), under the forward strut, at Z 10 or lower under the seat base, and up to the hull side only outboard of the seat edge. Harness aft: through the gland in the dry box aft wall, then in the 2 in conduit on rib standoffs, centerline at Z 4. Support every 18 in. Use lined metal clamps near the engine, belt and prop and along the cage. Stow gear beside the conduit, not on it.
9. Harness forward: dash run to starboard, bow run to port. Keep wires 9 in from dry exhaust unless there is a thermal barrier, and never run them directly above it.
10. TB1 and TB2 are screw bars: one ring terminal on each screw.
11. Side lights: red to port, green to starboard, on the hull sides at X 108, Z 17. Fit a wedge so each stands vertical on the flared side, and use a sealed gland through the hull.
12. Low oil light: connect W60, W61 and W62. First confirm from the service manual that the sensor closes on low oil.
13. Engine side: J1 connector, DESS and lanyard kill, tach, temperature sender, exhaust probe. Confirm everything marked "confirm" on the schematic from the service manual.

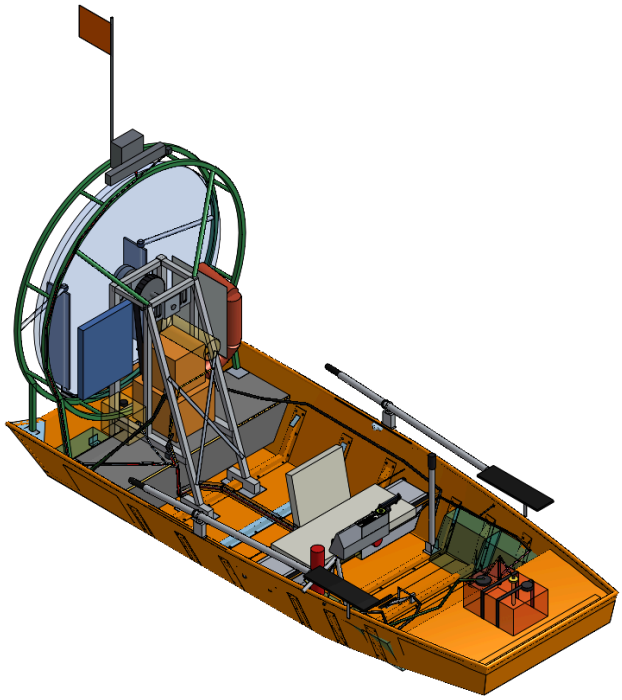
Engine systems:

14. Fuel line along the starboard side, about 1 in inboard of the skin, all joints clamped. Keep it away from battery cables.
15. Oil: connect the tank to the pump, fill it, and bleed the line exactly as the service manual says. This engine has no oil in the gas, so a bad bleed can destroy it. Run straight gasoline; never run the tank dry.
16. Cooling: connect the hoses and thermostat, fill and bleed. Trapped air here overheats the engine.
17. Tailpipe to the port transom corner: outlet at the transom top or higher, pointing aft and slightly outboard, never into the prop. Fit the pipe guard where a hand or knee could touch it.

Throttle:

18. Pedal, 3/16 in plate, at Y -2.5, hinged at the heel, with the rib at X 108 as its stop. Heel rests held by the UHMW screws.
19. Throttle cable from the post on the rib at X 108, along the starboard side to the engine, with a return spring at the pedal as well as at the throttle body. Confirm full open and full close. The engine must return to idle with your foot off.
20. Level and load label: bend the shelf from 0.063 in sheet and rivet it to the top of the pod. Fit the 2-3/4 in round bubble level at the port end and stick the load label beside it, both where you can read them from the seat. Set the level with the boat on its trailer, level side to side and about 1 deg stern down on the hull floor (use a digital angle finder), with a shim about 0.04 in thick under the aft edge of the level, so a centered bubble means the design trim. The step image does not show the level: see renderedImages/studio/v8_dash_driver-pov.png. Check it again afloat with a normal load. Read it at rest, because acceleration moves the bubble. Label text: MAXIMUM LOAD 380 LB. PEOPLE + GEAR + FUEL. FULL TANK 45 LB (EMPTY 8 LB) LEAVES 335 LB. KEEPS YOU WITHIN THE 825 LB CAPACITY PLATE. SHIFT GEAR TO CENTER BUBBLE. TIE DOWN.

Before moving on, check each circuit with a meter before fuses go in, then test the lights, gauges, bilge pump and low oil light. The starter stays disconnected until 2.14.



Skills: Marine wiring and crimping; Fuel system basics; Cooling system; Cable routing; Working from a schematic

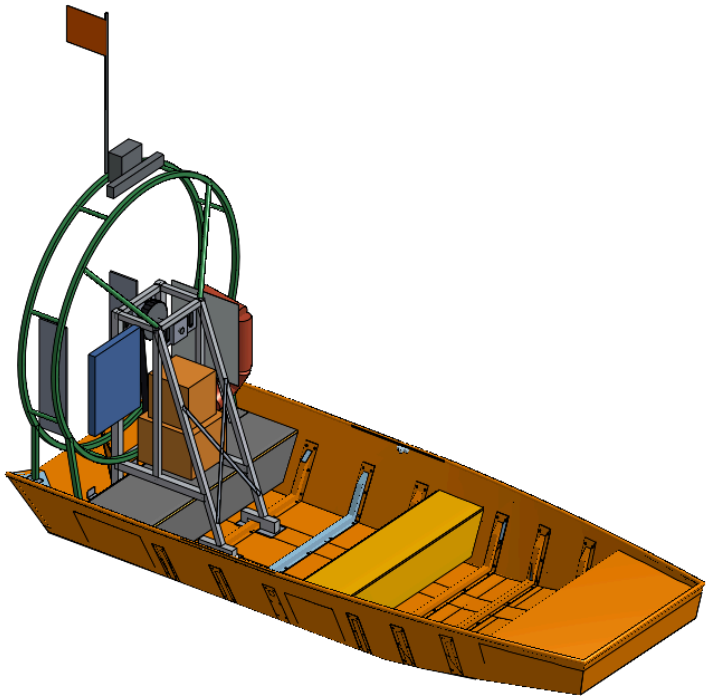
2.12 Build the prop cage around the prop position (boat)

	Materials	DIY hours	Pro hours	Pro cost if hired	Total if hired
Step 2.12	$620	16	8	$680	$1,300

Depends on: 2.7, 2.10 | Alongside: 2.11, 2.13

Goal: a cage built to the real prop position, with a way to fit the prop last.

1. Make a prop template: a 3/4 in plywood disc, 66 in diameter, bolted to the hub in place of the prop. Turn it by hand.
2. Build two hoops, 36 in radius, at X 1 and X 11, four legs at Y +/-30.5 on the pads from 2.7, and eight stringers.
3. Fit the two 1 x 0.065 in struts from the forward hoop to the top of the stand at X 21. Use rigid struts, never wires: a failed wire ahead of the disc can be drawn into the prop. Add rubber isolators at the strut ends if vibration calls for them.
4. Check clearance with the template: at least 2 in all round, and at least 2 in to the stand and bearing plates. The aft bearing corners and the aft hoop are tight, so check them against the real hub.
5. Make one section removable with bolts, big enough to pass the prop through. Label it.
6. Fit the mesh guard on the front and rim only, never in the prop wake: mesh there costs about 220 lbf of thrust. The cage is not a step. Remove the template.
7. Cage top: 20 in LED bar, searchlight and all-round white light. Connect their wires.
8. Flag: orange 10 x 12 in flag on the fold-down fiberglass whip, on the aft side of the cage, at least 10 ft above the bottom. Check the current FWC rule.



Skills: Mechanical assembly; Measuring; Tube bending or a bending shop; Bolting

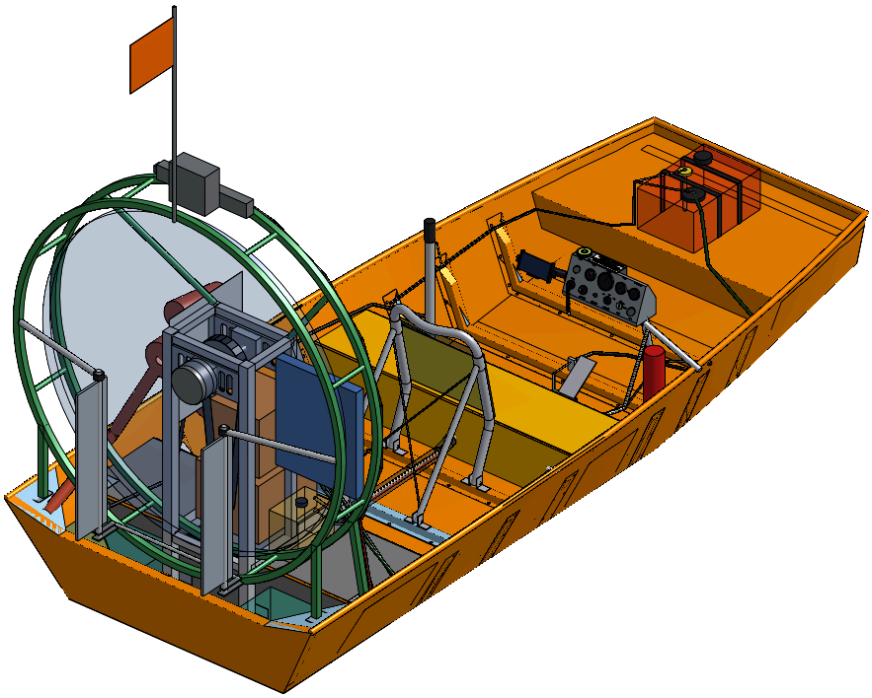
2.13 Install the stick, rudders and steering linkage

	Materials	DIY hours	Pro hours	Pro cost if hired	Total if hired
Step 2.13	$390	8	4	$340	$730

Depends on: 2.12 | Alongside: 2.11

1. Stick: base plate on the rib at X 95.25, Y 21, with the pivot cheeks. Steering stick, 1-1/2 x 0.120 in, and grip. Fit a stop on the cheeks that takes a hard pull before the rudder cable does.
2. Rudders: 8 x 30 in aluminum honeycomb blades (0.020 in skins over 1/2 in hex core; buy panel and cut it, close the edges with bonded aluminum channel, and pot solid hard points where the shaft and horn bolt through), hinged at 25% chord, on 1 x 0.125 in 6061 shafts. A 3/4 in shaft fails the load check, so don't downsize. Lower bracket on the aft hoop, upper arm out to the hoop.
3. Linkage: horn on each shaft, tie bar at X 0, and the 4.3 in outboard arm on the port shaft.
4. Cable: about 14 ft, along the port side, across the top of the rear bench to Y 18, under the hoops, to a clamp on the aft hoop.
5. Direction: stick forward turns left. Swap the cable to the other side of the rudder shaft to reverse it, and pick the habit before first running.
6. Set rudder stops at about 25 degrees each way, with equal travel and no binding. Check the tie bar at full rudder: it clears the hoop by only about 0.13 in at center.
7. Cotter pins or safety wire on every pivot.
8. Trim tab: fit the small tab and turnbuckle on the transom, to starboard (BOM line 79), with sealed bolts and a backing plate. Start it neutral.
9. Weigh before fitting: the two rudders with shafts, arms, brackets, horns and tie bar (allowance 8.5 lb, itemized about 7.8 lb), and the stick, pedal, cables and heel rests (allowance 8.5 lb, itemized about 6.1

lb). Write the weights in CAD/analysis/minimal_1448.js. If either set is over its allowance, the load label must come down.



Skills: Mechanical assembly; Linkage adjustment

2.14 Pre-test inspection and sign-off

	Materials	DIY hours	Pro hours	Pro cost if hired	Total if hired
Step 2.14	$0	2	1	$85	$85

Depends on: 2.11, 2.12, 2.13, 2.16, 2.17, 2.18 | Alongside: -

1. Work through Part A of reports/Static_Test_Protocol_Drivetrain.md.
2. Re-torque all structural bolts and check every torque stripe.
3. Nothing loose inside the cage or the boat. Flotation foam strapped. Seat bolted down, storage lids latched, seatback stop in. Oars shipped and strapped. Fluids full, no leaks. Belt tension checked.
4. Oil line bled, coolant bled, low oil light tested.
5. Connect the starter. Turn the engine by hand first to confirm the prop's rotation.
6. Two people sign the checklist.

Skills: Inspection; Torque wrench use

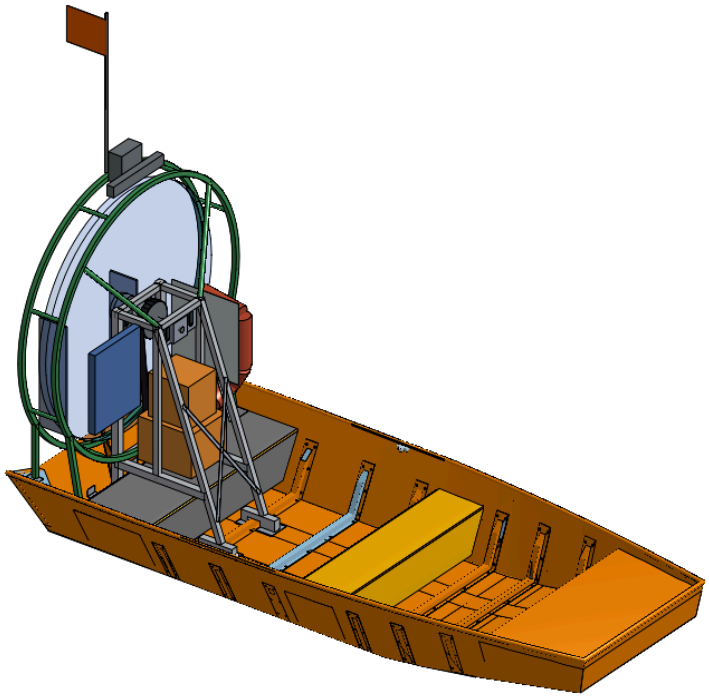
2.15 Install the propeller (last step before testing)

	Materials	DIY hours	Pro hours	Pro cost if hired	Total if hired
Step 2.15	$1,625	3	1.5	$128	$1,752

Depends on: 2.14 | Alongside: -

1. Lock out: key out, lanyard off, battery switch off.
2. Remove the cage section from 2.12.

3. Inspect the prop for damage. Check static balance and blade track, within 1/16 in.
4. Mount it with the face plate and the maker's bolts. Warp Drive HP: 3/8 in mounting bolts to 30 ft-lb in 5 ft-lb steps; 1/4 in blade clamping bolts to 10 ft-lb in 2 ft-lb steps, torqued at the nuts. Star pattern. Torque stripe or safety wire.
5. Turn it by hand. Check at least 2 in of tip clearance all round, and that nothing touches.
6. Reinstall the cage section.
7. Set the blade pitch on a tied-down run so the engine reaches 7,400 to 7,500 rpm at full throttle, never over 7,900 engine rpm or 3,200 prop rpm. The maker recommends static rpm 400 to 500 under the engine maximum. Adjust in 1 degree steps, within 6 to 18 degrees.
8. Re-torque the prop bolts after the first run, then check them at 5 hours, at 50 hours, and at least yearly. Limit full throttle tied to the dock to about 1 minute at a time, and watch the temperature.



Skills: Mechanical assembly; Torque wrench use; Prop balancing and tracking

2.16 Fit the flotation foam

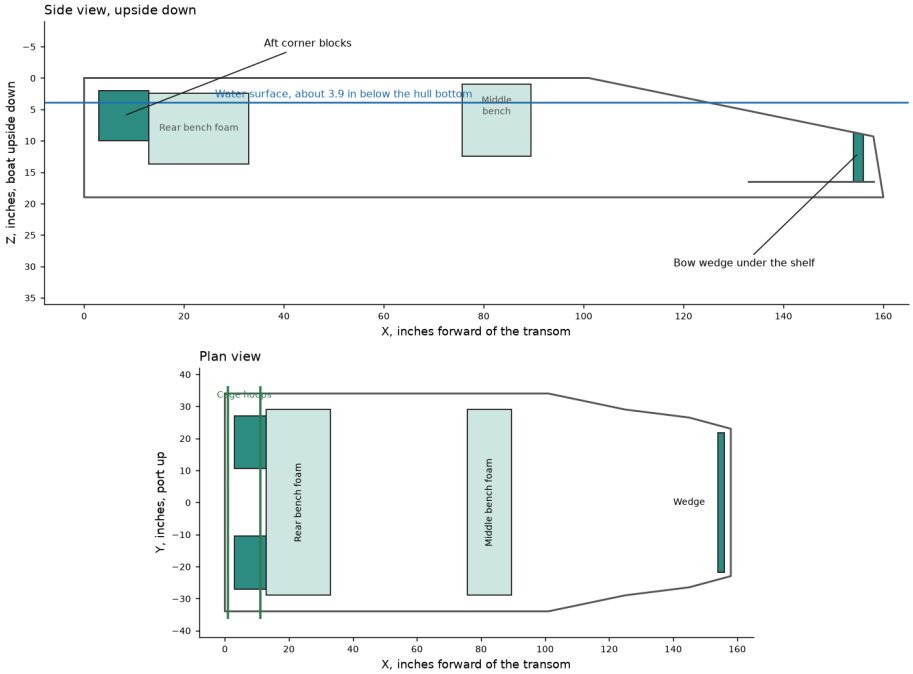
	Materials	DIY hours	Pro hours	Pro cost if hired	Total if hired
Step 2.16	$170	4	1.5	$128	$298

Depends on: 2.13 | Alongside: -

Foam so the boat floats level upside down. Analysis: CAD/analysis/flotation_inverted.js (v1.0.1.3). It relies on the rear bench being full of sound foam: check it first. The middle bench is gone; its flotation is replaced by the seat compartment floors and the closed-cell seat pad (both fitted in 2.17) and the two footwell side blocks below.

1. Rear bench foam: keep it intact. Cut only what steps 2.5 and 2.6 need, and seal the cut faces.
2. Foam: closed-cell, 2 lb/ft3, fuel resistant: polyethylene, or XPS sealed in a fuel-proof skin. Not beadboard.
3. Bow wedge: 2 x 43.6 in, about 7.5 to 7.9 in deep, at the forward end of the space under the front shelf (X 154-156). Cut it to fit the hull bottom.

4. Aft corner blocks, two the same: 10 in long (X 3-13), from Y 10.5 out to the hull side, Z 2 to 10. About 14.4 in wide at the bottom and 18.4 in at the top. Cut them to fit the hull side.
5. Port block: put a 0.040 in aluminum shield between it and the tailpipe above, or trim it to keep 9 in from the pipe.
6. Footwell side blocks: 6 in thick against each hull side, Z 1.5 to 13, from X 98 to X 128, forward of the stick base. Leave a gap in the starboard block at X 106-110 for the console foot. Cut each to the hull side. The fire extinguisher stands on top of the starboard block.
7. Strap all the blocks in place with stainless or aluminum hardware. Keep them clear of the bilge pump and hose, and leave a drain path under them.
8. When the boat is finished, roll it in shallow water with a crane or helpers. It should float level with the hull bottom about 4 in clear (the seatback foam counts: keep it closed-cell and attached). Trim the wedge to suit.



Skills: Measuring; Foam cutting; Mechanical assembly

2.17 Build and install the seat module: struts, electrical dry box, storage compartments, seat and seatback (boat)

	Materials	DIY hours	Pro hours	Pro cost if hired	Total if hired
Step 2.17	$700	19	9.5	$808	$1,508

Depends on: 2.5 | Alongside: 2.2, 2.3, 2.4, 2.8

Goal: the seat module where the middle bench was. Do it right after 2.5: keep the gunwale spreader in until the struts are bolted, and 2.11 needs the dry box. CAD: CAD/FeatureScript/Minimal_cockpit.fs (seat module).

Struts (the new thwart):

1. Two 1 x 2 x 1/8 in roll-wrapped carbon tubes, top at Z 11 (the old bench-top level), at X 76.5-77.5 and X 85.5-86.5. Cut each to the measured inside width less about 3/4 in (about 58.8 in) with a diamond blade, cutting wet or with a vacuum at the blade (P100 respirator). Seal the cut ends with epoxy. Wrap each tube

in one ply of fiberglass cloth and epoxy: it keeps the carbon off the aluminum. Drill every hole with a carbide bit and a backing block, and bond a 6061 crush sleeve into each one.

2. End clips: 2 x 2 x 1/8 in angle, 1 in wide, opened to the side flare. Bolt each to the hull side with two 1/4-20 stainless bolts, fender washers inside and sealant (reuse the bench holes where they line up), and to the strut through its crush sleeves, with a G10 shim between clip and strut and Tef-Gel on the stainless bolts: no bare carbon may touch aluminum. Take the spreader out only when all four clips are tight.

Electrical dry box:

3. Fold 0.063 in 5052 to 8 x 10 x 7-1/4 in outside (X 77.5-85.5, Y +/-5, Z 3.75-11). Rivet and seal the corners. Rivet a 3/4 in angle flange around the top with eight #10-24 rivnuts. Bolt the box flange to both struts with eight #10 stainless bolts in sleeves, with a G10 shim and Tef-Gel (no aluminum rivets into the carbon).
4. Lid: 0.080 in 5052, 8-1/2 x 10-1/2 in, closed-cell neoprene gasket, eight #10-24 stainless bolts.
5. Inside: battery pad and strap at the forward end; a bracket that holds the fuse block over the battery with the fuses facing the lid, so they can be tested and changed with only the lid off; a strip for relays K1-K5 at the aft end. Cable glands in the walls, the harness gland in the aft wall at about Z 5.3.
6. Battery switch plate: 1/8 in 6061, bolted to the forward strut through sleeves on a G10 shim, port of the box at about Y 8.

Storage compartments:

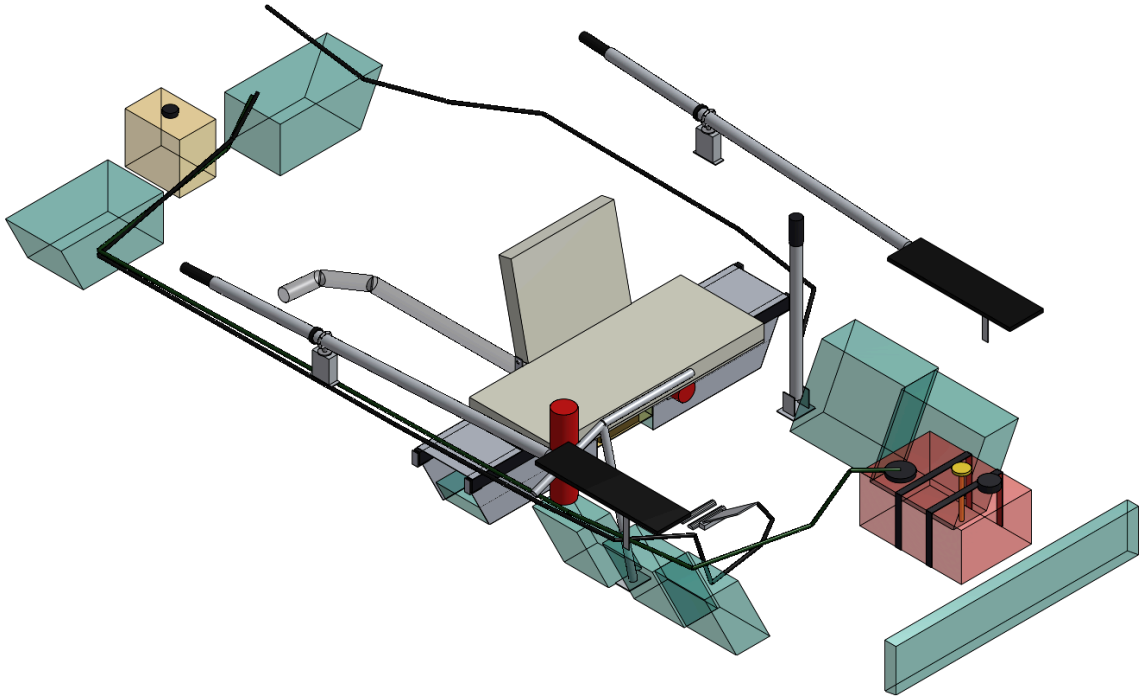
7. Walls: 0.032 in 5052, bolted under the struts with #10 stainless bolts on a G10 strip (not riveted to the carbon), from the box to the hull side, stopping 1-1/2 in above the floor so bilge water runs underneath. Seal the gap at the hull side with closed-cell foam tape; do not fasten the walls to the skin.
8. Floors: closed-cell polyethylene foam from Z 1.5 to Z 5, cut to the hull side. They are flotation (2.16) as well as the storage floors.
9. Lids: 0.063 in 5052, from about Y 19.5 out to 1 in short of the hull side, piano hinge on the aft strut, gasket, two latches each. They sit outboard of the seat and need not span the whole compartment.

Seat:

10. Base: 3/8 in fiberglass/foam sandwich, 15 x 38 in (X 76.5-91.5, Y +/-19): 5/16 in PVC foam core (about 5 lb/ft3) with about 0.020 in E-glass skins each side (one ply of 12 oz biaxial). Lay it up flat on a waxed table under a vacuum bag or weights, or buy a ready-made panel. At each of the four bolt holes, drill oversize, dig out the core and pot the hole solid with thickened epoxy, then drill to size. Seal all edges. It sits on closed-cell strips on the struts and is bolted down with four 1/4-20 bolts through sleeves in the struts. It covers the dry box lid: unbolt the seat to reach the fuses.
11. Pad: 2-1/2 in closed-cell polyethylene, glued to the base, covered in marine vinyl. It counts as flotation, so do not use open-cell foam.

Seatback:

12. Brackets: two 1/8 x 1-1/4 in 6061 plates bolted through sleeves to the aft face of the aft strut, on G10 shims, at Y +/-9.5, with a 3/8 in stainless pivot about 1/2 in above the seat foam (about X 75.9, Z 14.6) and a stop bolt that holds the back at 12 deg recline.
13. Back: 18 x 15 in, 1/4 in marine plywood, 1-1/2 in closed-cell pad on the front and 1/2 in on the back, marine vinyl. It folds flat forward onto the seat. Folded, you can sit facing aft on its back pad: engine off only, because you face the prop cage. This is the rowing seat (2.18).
14. Check: two people side by side, the stick at full travel clears the seat corner (about 1 in), and the driver's knees clear the console.



Skills: Measuring; Cutting and bending sheet aluminum; Riveting; Rivnuts; Drilling; Cutting and drilling carbon; Fiberglass layup and epoxy potting; Foam cutting; Upholstery

2.18 Fit the oarlocks and oar rests; set up and ship the oars (boat)

	Materials	DIY hours	Pro hours	Pro cost if hired	Total if hired
Step 2.18	$205	3	1.5	$128	$332

Depends on: 2.17 | Alongside: 2.16

Goal: row facing aft on the folded seatback with 7 ft oars, and carry the oars in their locks along inside each rail. Clearances: CAD/analysis/oars.js. CAD: CAD/FeatureScript/Minimal_cockpit.fs (oars).

1. Risers: 1-1/2 x 3 x 4 in 6061, pocketed to about 1/8 in walls, drilled for the oarlock socket. Centre each on the rail at X 66 (10-1/2 in aft of the seat's aft edge). Check the spot first: rudder cable to port, throttle cable and fuel line to starboard run about 4 in below the rail.
2. Bolt each riser through the rail with two 5/16-18 stainless bolts 2-1/4 in apart, a 1/8 x 1-1/2 x 4 in doubler under the rail lip, and sealant. Top-mount socket on the riser; the pin ends up about 9 in above the folded seatback.
3. Closed-ring oarlocks in the sockets. Fit each oar through its ring and set the rubber stop 24 in from the handle end.
4. Oar rests: a 1/8 in 6061 clip on each rail at X 118, with a padded cradle at the height of the ring (about Z 25) and a rubber strap.
5. Ship: swing each oar blade forward in its ring, lay it in the rest and strap it. The starboard oar passes about 4 in above the console arm where it meets the gunwale; the port oar about 10 in outboard of the stick. Neither touches the storage lids opened, the dash, the extinguisher or the sidelights. The handles end at about X 42, clear of the stand.
6. Check on the water, engine off: sit facing aft on the folded seatback, feet braced aft; the hands pass beside the knees, and the elbows clear the stick (about 5 in, stick at neutral: tie it there) and the dash at the finish.

7. Never row or ship the oars with the engine running: unstrap only with the engine off. Strap both oars before starting.



Skills: Measuring; Drilling; Machining aluminum (pockets); Bolting

Section 3 - Static Test

	Materials	DIY hours	Pro hours	Pro cost if hired	Total if hired
Section total	**$280**	**6**	**4**	**$340**	**$620**

3.1 Static test of the installed drivetrain

	Materials	DIY hours	Pro hours	Pro cost if hired	Total if hired
Step 3.1	$280	6	4	$340	$620

Depends on: 2.15 | Alongside: -

Follow reports/Static_Test_Protocol_Drivetrain.md step by step, with the crew roles and PPE from reports/Static_Test_Plan_Drivetrain.md.

Record data on the protocol data sheets, then fill in reports/Static_Test_Performance_Report.md.

Do not run the boat on the water until the static test has passed and the report is signed.

Skills: Following a test procedure; Engine operation; Data recording; Safety officer role (crew of 4)